NGUYEN VINH KHANH

Machine Learning Intern @ VPI | Data & AI Engineering | Analytics Engineering | BI & Product Analytics

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PROFESSIONAL SUMMARY

Final-year Data Science in Economics & Business student at National Economics University and Machine Learning Intern at the Vietnam Petroleum Institute. Builds production-oriented data systems across analytics engineering, business intelligence, forecasting, and applied AI. At VPI, owns a daily market-intelligence stack end to end: multi-source ingestion, PostgreSQL marts, grounded LLM reporting, a pre-write data-quality gate, and an internal dashboard portal. Portfolio spans RAG/LLM applications, a full MLOps serving stack, explainable market intelligence, satellite MRV, Power BI credit-risk analytics, and econometric research.

CORE SKILLS

Data & Analytics Engineering: Python, SQL, PostgreSQL, ETL/ELT, star-schema modeling, data marts, web/email ingestion, data-quality gates, point-in-time datasets, Excel

BI & Product Analytics: Power BI, Power Query, working DAX, Apache Superset, Streamlit, KPI design, dashboarding, cohort analysis, business reporting

AI / ML: scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, classification/regression, time series, ARIMA/GARCH/EGARCH, LLM/RAG, embeddings, ChromaDB, STT/TTS

Engineering & Cloud: FastAPI, pytest, Docker, MLflow, Prometheus/Grafana, Amazon Bedrock, AWS RDS, Google Earth Engine, Next.js, REST APIs

EXPERIENCE

Machine Learning Intern | Vietnam Petroleum Institute (VPI) - Digital Technology & Product Development — **2026 - Present**

- Own an end-to-end daily market-intelligence pipeline that ingests web sources and Outlook email attachments into AWS RDS PostgreSQL, generates grounded DOCX/PDF reports with Amazon Bedrock Nova, and serves dashboard-ready marts to Apache Superset.
- Consolidated a teammate's crawler and my own eight-source extractor into one four-stage orchestrator with per-stage failure isolation, freshness fallback so reports still ship on stale data, and single-shot Discord alerting; scheduled by Task Scheduler and a mail-arrival trigger.
- Built a pre-write data-quality gate, numeric hallucination guardrails on LLM synthesis, idempotent SQL migrations, and point-in-time known_at fields on a PostgreSQL instance shared with another engineer.
- Shipped an internal Next.js portal with Microsoft 365 sign-in and embedded Superset dashboards via guest tokens, plus a runbook and handover documentation for the full stack.
- Developed oil-price and volatility forecasting prototypes with walk-forward backtesting, GARCH-MIDAS and news indices; documented that price-level forecasts did not beat a naive baseline instead of overstating them.

Data / BI Intern | EZTECH — **Dec 2025 - Feb 2026**

- Built and maintained Power BI models for financial-statement analysis and P&L management, using Power Query to clean, transform, and integrate multi-source client data.
- Applied financial-reporting logic to KPI design and standardized inputs to improve reporting consistency, traceability, and accuracy.

SELECTED AI & ENGINEERING PROJECTS

RAG Translation & Interpreter Training Assistant — **In Development**

- Architecting a local-first AI assistant with a document RAG knowledge base, source-cited Q&A, a research agent, bilingual terminology extraction, mock interpreting simulation, STT/TTS, and feedback-based scoring calibration.
- FastAPI, ChromaDB, multilingual sentence-transformers, Gemini Flash, faster-whisper, SQLite, vanilla JS. Deliberately no LangChain so every external call passes one auditable egress layer that blocks on confidential documents; two-tier model routing falls back when a daily quota is hit.

Credit-Score Classification - End-to-End MLOps Platform | Course Project — **2026**

- Multiclass credit-score classifier benchmarking XGBoost, LightGBM, Random Forest and Logistic Regression, served through FastAPI with a web UI, tracked in MLflow with a model registry, and packaged with Docker Compose alongside Prometheus and Grafana.
- Added batch scoring, drift monitoring with alerting, CI/CD, and a four-tier test suite covering unit, integration, contract, and smoke tests.

InsightX Market Copilot | Market Intelligence MVP — **2026**

- Built an explainable market-intelligence pipeline and five-page Streamlit app for separate Vietnam/Indonesia EDA, product matching, human review, competitive indicators, and review-only pricing actions.
- Verified with 19 passing tests and 10/10 page-market smoke checks; human-reviewed 15 candidate pairs and supported action proposals for four accepted pairs, under explicit guardrails against cross-currency comparison and unverified margin claims.

MRV Carbon Monitoring Pipeline | Technical Portfolio Piece — **2026**

- Built Sentinel-1/2 acquisition, spectral and SAR feature extraction, and data-density reconnaissance on live Google Earth Engine data for a 2.03 km^2 rice-paddy pilot in Bac Ninh; delivered a credential-free Streamlit demo and 100 passing tests.
- Caught a cloud metric silently reading a perfect score on every scene; the fix deleted my own headline result and exposed that optical imagery was unusable in the rainy season, which is why Sentinel-1 SAR was added as an independent physical cross-check. Scope kept honest as crop-season phase detection, with no accuracy claimed anywhere.

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Selected Data, AI & Product Work

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GEOSPATIAL & RISK ENGINEERING

Climate Risk Layer for Agricultural Lending | Technical Portfolio Piece 2026

- Mapped Typhoon Yagi flood extent from Sentinel-1 SAR and overlaid it on cropland and administrative boundaries to expose correlated default risk that independence-assuming PD models miss; compared a financial-only model against a financial-plus-flood layer.
- Real satellite flood data over an explicitly labelled synthetic loan portfolio, with a standing rule never to claim default-prediction accuracy without ground truth; 37 unit tests.

ANALYTICS & RESEARCH PROJECTS

Loan Default Risk Dashboard | Datapot NextGen Analytics Challenge 2025 - Top 6 2025

- Built a five-page Power BI dashboard over a 255K-loan portfolio with 32.6bn exposure and an 11.6% default rate: portfolio overview, borrower profiling, risk-lift comparison across credit-score x DTI segments, and Key-Influencer analysis identifying low income (2.4x) and age under 25 (2.2x) as dominant drivers.
- Designed a credit-policy simulator that quantifies, per segment, loans cut, defaults avoided, exposure reduced, and estimated expected loss saved, turning descriptive risk analysis into a decision tool.

Silent Exit - E-commerce Churn Analytics | Course Research, Individually Extended 2026

- Built an 11-module churn pipeline on a synthetic 8,000-customer, 25,858-order dataset: recency-based labelling, 29 RFM and behavioural features, Logistic Regression versus Random Forest with SHAP attribution, revenue and CLV at risk, and a 664-customer Priority Save Zone.
- Reported the discount-trap hypothesis as unsupported and documented label-feature leakage as a known limitation rather than quoting the inflated headline AUC.

Geopolitical Risk & Vietnamese Gold Volatility | Individual Paper Extending Group Research 2026

- Modelled 2010-2025 Vietnamese gold returns and volatility with ARX-EGARCHX, placing the Caldara-Iacoviello Geopolitical Risk index in both mean and variance equations and separating positive from negative shocks to test asymmetry.
- Replaced the group paper's researcher-chosen sub-periods with Bai-Perron endogenous break detection and 250-day rolling-window coefficient tracking; robustness via quantile regression, Ljung-Box and ARCH-LM diagnostics.

Consumer WTP for Regenerative Rice | Ongoing Scientific Research, NEU 2026 - Present

- Designing a randomized information-provision experiment in Hanoi to causally measure how carbon-framed versus health-framed messages shift willingness to pay for regeneratively farmed rice, built on Theory of Planned Behaviour and Theory of Consumption Values with greenwash-scepticism controls.

PRODUCT, STRATEGY & FINANCIAL ANALYTICS

CampusCalm | TECHSEED 2025 - Top 4 | Team Lead, Technology 2025 - 2026

- Led the technical build of an AI mental-health companion platform for Vietnamese students: FastAPI backend, Next.js web app, and an Expo mobile client; reached the TECHSEED 2025 final round.

AI Debt Untangle | AI-assisted debt planning and financial wellness 2026

- Designed the v2 SQLite data model separating actual cash flows from what-if projections, adding a scenario layer for the optimizer and a cohort table wiring cited secondary data with URL-level provenance; co-developed the B2B2C strategy, KPI/GTM roadmap, pricing, and privacy principles.

Capital X (XCAP) - SME Revenue-Based Financing Marketplace | XPLORATORS 2026 - Top 5 2026

- Co-designed a marketplace pooling SME revenue-based financing into a three-tranche risk structure; built unit economics, investor segmentation, risk architecture, and phased go-to-market logic.

AWS AI Credit Engine for SME Lending | Competition Project 2025 - 2026

- Designed a managed AI-credit platform concept combining cash-flow and alternative data, fraud controls, human review, cost analysis, and a bank-pilot deployment roadmap.

Banh Oan Hoa Sen | Co-founder & Product Manager | F&B venture 2025

- Defined the MVP and ran the numbers myself: unit economics (COGS, margin, break-even), pricing, and an order and inventory tracker. Reached roughly 100 units sold and VND 30M revenue within three months of launch, and documented SOPs so the process survived being handed over.

EDUCATION & ACHIEVEMENTS

B.Sc. Data Science in Economics & Business | NEU Expected Feb 2027

- GPA: 3.27/4.0. Coursework: Probability, Statistics, Econometrics, Time Series Analysis, Machine Learning, MLOps, Data Preparation & Visualization, Database Management, Data Structures & Algorithms.

Top 4 TECHSEED 2025 (CampusCalm) | Top 5 XPLORATORS 2026 | Top 6 Datapot NextGen Analytics Challenge 2025 | Top 25 UEB Data Showdown

Vice Head of Project Department, Startup Innovation Dynamic NEU | Co-founder & Head of Specialist Department, Chess Elite NEU

Chess instructor at three Hanoi primary schools since 2024 (classes of 15-50, ages 6-10) and 1-on-1 mathematics tutor since 2023 | 16 years competitive chess, national youth medallist

Vietnamese (native) | English (B2 CEFR) | Microsoft Office Specialist certified (Word, Excel, PowerPoint)